

Data Flow Diagram (DFD Specification)

VANGUARD CREDIT AUTOMATION | ACADEMIC PROJECT TRACK DELIVERABLE

1. DFD Level 0 (Context Level)

The Web Browser interacts with the Flask Application Server via HTTPS. Inputs are sent via JSON payloads, and predictions are returned as JSON response sets.

2. DFD Level 1 (Application Flow)

- Step 1: User fills out credit variables on the dashboard.
- Step 2: Input variables are structured into an HTTP POST request payload.
- Step 3: Flask endpoint captures the payload and loads model/preprocessor objects.
- Step 4: Numeric variables are scaled using StandardScaler; categorical variables are mapped.
- Step 5: Model predicts class label (0/1) and class probabilities.
- Step 6: Rule overrides are applied for high-risk credit checks.
- Step 7: Prediction response is serialized into JSON and returned to render UI widgets.